AI-READABLE KNOWLEDGE BASE

Department Reference Document

Department of Mechanical Engineering (MECH)

1. Department Overview

Mechanical Engineering is the broadest engineering field with ample application areas. It encompasses an enormous range of technical disciplines including Refrigeration & Air-conditioning, Control & Monitoring Mechanisms, Computer-aided Design, Energy Management, Fluid Mechanics, Tribology, Robotics, Biomechanics, Turbomachinery, Materials and Nanotechnology, and Mechatronics. In the ever-changing scenario of present-day technology, Mechanical Engineering is a challenging, rewarding, and highly respected profession.

The Department of Mechanical Engineering was established in the year 2009 with an initial intake of 60 students. The department also offers an M.Tech program in Thermal Engineering with an annual intake of 18 students.

Accreditation

- The B.Tech (Mechanical Engineering) Program has been accredited by the National Board of Accreditation (NBA) since 2016 for 3 years.
- In 2019, the program received NBA TIER-I accreditation for 3 years.

Department Head

Name: Dr. J. Krishnaraj

Experience: Vast teaching experience

Faculty and Staff

Total Faculty: 30 well-experienced and qualified faculty members

Technical Staff: 6

Faculty with Ph.D.: 6, with research, academic, consultancy, and industrial experience

Faculty Research Areas

- Alternate Fuels and IC Engines
- Composites and Biomedical Applications
- Manufacturing and Welding Technology
- Robotics and CAD/CAM

Consultancy Achievements

The department has demonstrated consultancy abilities through design and fabrication works of sports and structural equipment, including work for a badminton academy in Hyderabad.

2. Vision and Mission

Vision

The Mechanical Engineering Department endeavors to be recognized globally for outstanding education and research, leading to well-qualified engineers who are innovative, entrepreneurial, and successful in advanced fields of mechanical engineering to cater to ever-changing industrial demands.

Mission Statements

- Impart highest quality education to students to build their capacity, enhance their skills, and make them globally competitive mechanical engineers and successful entrepreneurs.
- Provide students with an academic environment of excellence, state-of-the-art research facilities, leadership, ethical guidelines, and lifelong learning needed for a long and productive career.

3. Programs Offered

B.Tech - Mechanical Engineering

Year Established: 2009

Initial Intake: 60 students

Accreditation: NBA accredited since 2016; NBA TIER-I accreditation received in 2019

M.Tech - Thermal Engineering

Annual Intake: 18 students

4. Program Outcomes (POs)

Engineering Graduates of the Mechanical Engineering program will be able to:

PO1. Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.

PO4. Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

PO5. Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The Engineer and Society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

PO7. Environment and Sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for, sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, including writing effective reports, design documentation, presentations, and clear instructions.

PO11. Project Management and Finance: Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work as a member and leader in a team to manage projects in multidisciplinary environments.

PO12. Life-long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO1: To solve complex mechanical engineering problems, enabling students to be employable in reputed industries or to become self-entrepreneurs.

PSO2: To develop new products/systems in collaboration with research and development centres, and to continue higher education.

5. Infrastructure and Facilities

Classrooms

The college has state-of-the-art classroom facilities for all classes of Mechanical Engineering, equipped with smart boards. A seminar hall with the latest LCD projector is available for various seminar activities.

Seminar Hall

A well-furnished seminar hall caters to departmental needs such as presentations of mini and main projects and other special programs.

Department Library

A departmental library has been set up to cater to the requirements of the department.

6. Laboratories

Engineering Workshop Lab

Purpose: Provides the area and tools required for manufacture or repair of manufactured goods, with all basic requirements of components.

Used By: Mechanical Engineering students for hands-on fabrication and repair activities.

Fluid Mechanics and Hydraulic Machinery Lab

Purpose: Students understand various methods of measurement of flow rates, capacity, and velocity of water in tanks, closed pipes, and open channels. Students also gain experience understanding fluid properties like surface tension, temperature, and viscosity.

Facilities: Range of experimental set-ups for flow measurement and fluid property determination; useful for design of drainage channels and open channels.

Used By: Mechanical Engineering students studying fluid mechanics and pump selection.

CAD/CAM Lab (Computer-Aided Design and Computer-Aided Manufacturing Lab)

Purpose: Gain practical experience in 2D drafting and 3D modeling; exposes students to modern control systems (Fanuc, Siemens, etc.).

Facilities: Computers (i3 processor, 4 GB RAM, Standard Graphics Card, 1TB Hard Disk); CNC Machine features; AUTODESK certification programs; advanced CAD/CAM/CAE/CFD software.

Used By: Mechanical Engineering students and faculty for professional certifications, industry consultancy, and research.

Heat Transfer Lab

Purpose: Students understand all three modes of heat transfer: conduction, convection, and radiation through practical experiments.

Facilities: Experimental setups for conduction, convection, and radiation studies; starting points for higher-level experimental heat transfer studies.

Used By: Mechanical Engineering students studying heat transfer theory and applications.

Thermal Engineering Lab

Purpose: Show students experimental methods on thermal energies on various engines and demonstrate their operational procedures; values are used to determine fuel properties.

Facilities: Experimental setups for thermal energy analysis on different engines; classroom lectures precede lab work for thorough theoretical understanding.

Used By: Mechanical Engineering students studying thermal systems and IC engines.

Machine Tools Lab

Purpose: Introduce students to the know-how of common manufacturing processes involving material removal. Auxiliary methods for machining to desired accuracy are covered.

Facilities: Equipment for understanding basic features of manufacturing processes; focus on process understanding rather than machine construction details.

Used By: Mechanical Engineering students studying manufacturing processes.

Production Technology Lab

Purpose: Covers Metal Casting, Welding, Press Working, and Processing of Plastics. Inculcates knowledge and skill from preparing wooden patterns to completion of castings, including Sand Testing techniques.

Facilities: Equipment for metal casting, welding, press working, plastics processing, and sand testing; covers different manufacturing processes used in industry.

Used By: Mechanical Engineering students studying manufacturing and production processes.

Mechanics of Solids Lab

Purpose: Students apply loads to various materials under different equilibrium conditions and perform tests in tension, compression, torsion, bending, and impact.

Facilities: Universal testing machines, torsion equipment, spring testing machine, compression testing machine, impact tester, hardness tester, dial indicators, extensometers, strain gauges, strain indicator equipment, and graphing capabilities.

Used By: Mechanical Engineering students studying strength of materials and material behavior under loading.

Instrumentation and Control Systems Lab

Purpose: Students measure and calibrate physical quantities like pressure, temperature, speed, displacement, flow, and vibration in real-time engineering applications using transducers with amplification circuits.

Facilities: Instruments for measuring flow, temperature, level, distance, angle, and pressure; supports both remote and automated control capabilities.

Used By: Mechanical Engineering students studying measurement, monitoring, and control systems.

Dynamics of Machines Lab

Purpose: Students understand and analyze the balancing of machine parts statically and dynamically; analyze dynamic behavior of moving objects; analyze and synthesize machines for mobility and forces.

Used By: Mechanical Engineering students studying machine dynamics and kinematics.

Metallurgy and Material Science Lab

Purpose: Study of physical and chemical properties of metallic elements and their alloys.

Facilities: High-tech equipment for failure testing, corrosion testing, chemical analysis, fatigue testing, metallography, and weld testing.

Used By: Mechanical Engineering students studying materials science and metallurgical properties.

Metrology and Measurement Lab

Purpose: Equip students with knowledge of common linear and angular measuring instruments for engineering measurements.

Facilities: Instruments for linear and angular measurement; follows the principle that measurement is the process of experimentally obtaining quantity values attributed to a property of a body or substance.

Used By: Mechanical Engineering students studying precision measurement and quality control.

Composites Lab

Purpose: Manufacturing, characterization, and modeling of advanced composites and nanostructured materials; preparation of different combinations of composite and hybrid materials.

Facilities: Furnace, Bottom Pouring Type Stir Casting Machine, Digital Ultrasonic Flow Detector, Barcol Hardness Tester.

Used By: Mechanical Engineering students and faculty working on composite materials research and fabrication of models for research projects.

7. Centres of Excellence (COEs)

Centre of Excellence - Digital Manufacturing

Partner: Internal departmental centre

Objective: To provide the basic and advanced features of manufacturing techniques such as CNC machining and additive manufacturing, and to motivate students and faculty to undertake research and consultancy projects in digital manufacturing.

Outcomes

- Students will be able to model any 3D component requiring precision manufacturing in Mechanical and Aerospace engineering.
- Students will become familiar with making complex components using digital manufacturing techniques required by industries.
- Research publications by students and faculty.
- Improved industry interaction through consultancy projects.

Scope

This COE can be extended to carry out consultancy work fulfilling the needs of small and medium-scale industries. Students can fabricate prototype models for their research projects.

Major Resources Available

- CNC Lathe
- CNC Drill Tap Machining Center (CNC Mill)
- 3D Printer (Ultimaker 2) with Cura software
- CADEM Software - 5 users

Centre of Excellence - Product Life Cycle Management (PLM)

Partner: TATA Technologies

Objective: To inspire sustainable thinking across the entire product lifecycle, representing an all-encompassing vision for managing data relating to design, production, support, and ultimate disposal of manufactured goods.

Outcomes

- Students are able to list, justify, and interpret productivity models in manufacturing and service organizations.

- Understanding of manufacturing resources to predict and control the quality of an end product.

Scope

Provides knowledge about how a Product Lifecycle Management system (PLM) is used to structure and manage the information which guides the product during its lifecycle.

Major Resources Available

- ENOVIA by Dassault Systemes
- WINDCHILL by PTC

Centre of Excellence - Advanced Composite Materials

Objective: Focus on consultancy work for small and medium enterprises for industry-level projects; helps students understand various characteristics of composites and fabricate models for research projects.

Facilities: Furnace, Bottom Pouring Type Stir Casting Machine, Digital Ultrasonic Flow Detector, Barcol Hardness Tester.

Used By: Mechanical Engineering students for preparation of composite and hybrid material combinations.

Centre of Excellence - Computational Engineering by Dassault Systems

Software: CATIA V5R18

Activities: Computational projects in preparation, modeling, analyzing, and interpreting data. Computational results are validated with theoretical approaches in student projects. Staff utilize this facility for research work and projects presented at national and international conferences and published in international journals.

Centre of Excellence - Non-Destructive Testing

A dedicated Centre of Excellence for Non-Destructive Testing (NDT) is available in the department. Advanced NDT training programs are conducted in collaboration with Synergem Technologies Pvt Ltd.

Centre of Excellence - Advanced Welding Technologies

Partner: Synergem Technologies Pvt Ltd

A Centre of Excellence for Advanced Welding Technologies is set up in the department in collaboration with Synergem Technologies Pvt Ltd.

SAE (Society of Automotive Engineers) Workshop

Purpose: Fabrication and development of vehicles for various competitions.

Facilities: Windchill TIG Welding, Arc Welding, Lathe, Portable Metal Cutter, Drilling Machine.

Used By: II, III, and IV B.Tech (Mechanical Engineering) students through the SAE professional society.

8. Key Differentiators

The Department of Mechanical Engineering is unique in the following aspects:

- Industry Integrated Curriculum
- aSOP (Automotive Student Orientation Program) by Society of Automotive Engineers (SAE)
- Project-based learning
- Learning by doing approach
- Industry readiness training
- Encouragement of students to learn by practice
- Building automobile modules and complete vehicles by students of II, III, and IV B.Tech (MECH) through SAE professional society
- Value-added embedded training programs on geometric modeling, analysis, and digital/CNC manufacturing for selected students
- Innovative micro projects
- Extended laboratory facilities beyond college hours for interested students
- Technical seminars for better speaking skills for students from II B.Tech onwards
- MOOCs available to faculty members
- Research promotion schemes for faculty members
- Electronic drawing implementation in Engineering Graphics, Machine Drawing, and Production Drawing
- Student-centric teaching and learning methodologies

9. Software and Technology Tools

The department uses the following software tools for modeling, analysis, and manufacturing:

- ANSYS
- CATIA
- CREO
- ROKISIM
- PTC WINDCHILL
- AUTOCAD
- CADEM
- STABULI
- MATLAB
- ENOVIA

Key Machines and Equipment

- ACE Micromatic 6-Station Vertical Machining Centre
- CNC Turning Centre
- 3D Printing Technology
- Supporting CAD/CAM Laboratory

10. Training Programs

Mechanical Training Programs

Sl. No.	Name of Training	No. of Students	Company	Duration
1	Introduction to NDT in Aerospace and Mechanical Engineering	29	Synergem Technologies	02-02-2026 to 28-02-2026
2	Fundamentals of Material Testing and Evaluation	26	Synergem Technologies	01-09-2025 to 20-09-2025
3	Introduction to NDT in Aerospace and Mechanical Engineering	23	Synergem Technologies	03-11-2025 to 29-11-2025
4	Study of NDT Standards and Safety Procedures	23	Synergem Technologies	03-02-2025 to 01-03-2025
5	Industrial Training on Weld Inspection Using NDT Techniques	45	Synergem Technologies	02-09-2024 to 28-09-2024
6	Non-Destructive Evaluation of Composite Materials	46	Synergem Technologies	22-01-2024 to 10-02-2024

11. Industrial Visits

Sl. No.	Name of the Industry	Date of Visit	No. of Students
1	Mahindra & Mahindra	29-11-2025	46
2	Pennar Industries	31-01-2025	48
3	Mahindra & Mahindra	06-01-2024	52
4	Pennar Industries	18-08-2023	46

12. Activities

Faculty Activities

- TASL Employees Training - 2026
- SAE KRT Club Inauguration
- Guest Lecture by Prof. Venkateshwar Reddy at Anurag University - 2026
- Faculty Training on Fusion 360 Generative AI - 2024

Student Activities

- Session on Higher Education by Mr. Pawandeep Singh - 16-03-2026
- Drone Design Challenge at Chennai - February 2026
- Auto Desk Design Labs - Drive Innovation Forward: In-Demand 3D Design Skills - 2025
- National Level Student Convention on CAD - 2024
- National Level Competition on Reverse Engineering - 2023

13. Placements

Students of the department have been placed in various companies across industries including manufacturing, IT, automotive, energy, and consulting. Companies include:

- nDMatrix, TATA Steel, TASL, GE, Ashok Leyland, L&T Technology Services, Mahindra
- Pennar Industries, EIS, Verzeo Edu Tech, Schneider Electric, SAFRAN
- LTI, Hexaware, Grey Campus, CTC, KIA R&D (HYUNDAI), MG, KPIT Cummins
- Shakti Hormann, Raam Group, IXAR, Accenture, Cognizant, HCL Technologies
- People Link, MEIL, RDC Concrete, Maker Global, Multiplier Solutions, BYJU's
- Cyient, Tech Mahindra, Hyundai Transys, TAFE, Medha Servo Systems
- Wipro, TCS, Sterling & Wilson, Capgemini, Rane Engineering, Vem Technologies
- Deepak Nitrite, IBS Software, NTT Data, Infosys, Rapid Robotics, and many more

A significant number of graduating engineers are pursuing higher studies in various universities abroad by securing qualifying grades in GRE, TOEFL, and IELTS examinations.

End of Knowledge Base Document | Department of Mechanical Engineering